

# FLOURISH OLAIYA

AI & Machine Learning Engineer

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## PROFESSIONAL SUMMARY

AI & Machine Learning Engineer with 5+ years of experience designing, deploying, and maintaining production-grade intelligent systems, with strong expertise in Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and agentic workflow orchestration. Proven ability to translate early-stage research prototypes into scalable, secure, and maintainable production systems using modern MLOps and cloud-native architectures. Experienced in building traceable, reliable, and safety-aware AI solutions with strong emphasis on model evaluation, hallucination mitigation, and responsible AI deployment. Passionate about applying AI to fintech, healthcare, education, and safety-critical environments.

## CORE TECHNICAL SKILLS

|                        |                                                                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| LLMs & AI Systems      | LLM Integration & Agentic Systems, RAG, Multi-step Reasoning, Prompt Engineering & Guardrail Design, Model Evaluation & Output Traceability |
| ML & Deep Learning     | TensorFlow, PyTorch, Scikit-Learn, Keras, Hugging Face Transformers, Computer Vision (YOLO, OpenCV, ONNX) & Signal Processing               |
| LLM & AI APIs          | Gemini API, OpenAI API, Groq API, Hugging Face Inference Endpoints                                                                          |
| MLOps & Deployment     | FastAPI, Flask, Streamlit, Gradio, CI/CD Pipelines, GitHub Actions, Docker & Containerized Deployment, Model Monitoring                     |
| Vector Databases       | Pinecone, ChromaDB, Semantic & Hybrid Search (BM25), Knowledge Indexing                                                                     |
| Cloud & Infrastructure | AWS (S3, EC2, Lambda), Vertex AI, Render, Vercel, Hugging Face Spaces                                                                       |
| Data & Analytics       | PostgreSQL, MongoDB, Power BI, Excel, DAX                                                                                                   |
| AI Safety & Compliance | Responsible AI Practices, Bias Detection, Explainability (SHAP, LIME, Saliency Mapping), Secure AI & Data Privacy                           |

## PROFESSIONAL EXPERIENCE

### ML/AI Engineer — zKYTE Technologies

Jan 2026 – Present

- Designing and deploying machine learning solutions to support data-driven product features and business intelligence.
- Building and maintaining scalable ML pipelines for model training, evaluation, and production inference.
- Collaborating cross-functionally with engineering and product teams to integrate intelligent systems into core platforms.
- Implementing model monitoring, performance tracking, and continuous improvement workflows.
- Applying responsible AI principles including fairness evaluation, bias detection, and output traceability.

## AI Engineer — Schoolinka

Jun 2025 – Dec 2025 (Contract)

- Designed and deployed a production RAG-based conversational coaching system supporting WhatsApp integration with real-time personalized responses.
- Developed multi-agent LLM workflows integrating Gemini API, Hugging Face models, and Pinecone vector search for structured knowledge retrieval.
- Built scalable FastAPI microservices enabling modular AI orchestration, inference optimization, and low-latency user interaction.
- Implemented evaluation pipelines for monitoring LLM output quality, hallucination detection, and response reliability.
- Designed data privacy safeguards and secure user data processing pipelines aligned with responsible AI principles.

## Data Science Lead — Data Scientists Network (DSN FUNAAB)

Feb 2025 – Jan 2026 (Volunteer)

- Led and mentored 200+ members in applied machine learning, analytics engineering, and AI system design.
- Designed real-world predictive analytics solutions across education and business intelligence use cases.
- Developed training materials and workshops covering ML deployment, feature engineering, and data visualization.

## Community Lead — Robotics & AI Innovators Network

May 2024 – Apr 2025

- Coordinated applied robotics and AI engineering programs involving 150+ participants.
- Supervised R&D projects in automation, intelligent perception, and embedded AI.
- Led cross-functional teams delivering robotics and intelligent automation prototypes.

## Data Analyst Associate Intern — Excelerate

Jul 2024 – Aug 2024

- Performed large-scale data analysis to identify behavioral and operational trends.
- Developed executive dashboards using Power BI supporting data-driven decision-making.
- Delivered actionable analytics insights that improved stakeholder strategic planning.

## Data Science Intern — Hamoye

Jul 2022 – Sep 2022

- Developed regression and classification models for predictive analytics and sports performance forecasting.
- Performed feature engineering, hyperparameter tuning, and model performance optimization.
- Collaborated within agile data science teams to deploy reproducible ML pipelines.

## SELECTED AI & ML PROJECTS

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### Telecom Fraud & Anomaly Detection Pipeline

[github.com/lourish789/refactored-telecom-sniffle](https://github.com/lourish789/refactored-telecom-sniffle)

- Built an end-to-end ML system detecting SIM-box, IRSF, SIM-cloning, and subscription fraud from Call Detail Records using IsolationForest and XGBoost dual-model risk scoring.
- Developed a LangGraph-based AI investigation agent generating structured fraud case reports, served through FastAPI with a live monitoring dashboard and stream scoring.

### Real-Time Catfish Fingerling Counting System

[github.com/lourish789/fy\\_fingerling\\_project](https://github.com/lourish789/fy_fingerling_project)

- Developed a YOLOv8 and SORT-based computer vision system that detects, tracks, sorts, and counts fingerlings from live camera, video, or image streams in real time.
- Built an interactive web dashboard with live video streaming, virtual counting lines, class-based sorting, and CSV logging for hatchery operations.

### AI Coaching Assistant for Teachers

[github.com/lourish789/schoolinka](https://github.com/lourish789/schoolinka)

- Built an agentic RAG-powered conversational assistant deployed via WhatsApp with real-time response generation.
- Implemented semantic retrieval using Pinecone and structured knowledge orchestration for personalized coaching responses based on intent classification and multi-turn history.

### Multi-Modal Media Archive RAG System

[github.com/lourish789/RAG\\_system\\_berlin](https://github.com/lourish789/RAG_system_berlin)

- Built a production-grade retrieval system over audio interviews and PDF documents with Whisper transcription, speaker diarization, and hybrid semantic + BM25 search.
- Implemented strict source attribution with timestamps and page citations, plus automated faithfulness and relevance evaluation metrics.

## AgriTech Suite — AI Farming Platform

[github.com/lourish789/crispy-funicular](https://github.com/lourish789/crispy-funicular)

- Built a full-stack agricultural platform with on-device plant and livestock disease diagnosis from images or video using local Hugging Face vision models with LLM enrichment.
- Integrated a context-aware streaming RAG chatbot, personalized farm advisory engine, marketplace, and community hub with graceful offline fallbacks at every AI stage.

## Multi-Modal Breast Cancer Detection System

- Developed CNN-based diagnostic models for MRI, ultrasound, and mammography images achieving high cross-modal accuracy.
- Deployed inference applications via Hugging Face Spaces and Gradio with SHAP-based explainability visualizations supporting medical interpretability and clinical trust.

## Cancer Patient Support Chatbot

[github.com/lourish789/Simple-Whatsapp-Chatbot](https://github.com/lourish789/Simple-Whatsapp-Chatbot)

- Built a safety-aware WhatsApp RAG chatbot for cancer patients and caregivers with intent recognition, curated medical retrieval, automatic disclaimers, and privacy-focused local data storage.

## AI SAFETY, RELIABILITY & GOVERNANCE

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- Designed guardrails to reduce hallucination and adversarial prompt injection in production LLM systems.
- Built evaluation workflows for LLM factual consistency, response traceability, and output reliability scoring.
- Implemented explainable AI techniques (SHAP, LIME, Saliency Mapping) supporting transparency and interpretability.
- Applied privacy-preserving design and secure ML deployment principles across healthcare and education systems.

## EDUCATION

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### B.Eng Agricultural & Bio-Resources Engineering

2021 – Present

Federal University of Agriculture, Abeokuta (FUNAAB)

## CERTIFICATIONS

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- Google Data Analytics Professional Certificate
- Introduction to Machine Learning — Kaggle
- Intermediate Machine Learning — Kaggle
- Introduction to Deep Learning — Kaggle
- Computer Vision — Kaggle
- Data Science with Python — Udemy